PREPARED FOR PORTFOLIO REVIEW | CONFIDENTIAL

Retail Intelligence Platform Final Report

An end-to-end retail analytics platform combining descriptive dashboards, sales forecasting, customer segmentation, and product recommendations — with data-driven findings on where the business is losing margin and why.

Dataset: Sample Superstore (2014–2017) | 9,994 order records | 793 customers | 1,862 products
Tools: Python, Streamlit, Plotly, scikit-learn, SQLite, Power BI
Version 2.0

Table of Contents

1. Executive Summary **2**
2. Business Problem & Objectives **3**
3. Data Source & Methodology **4**
4. Key Findings: Sales & Profit **5**
5. Discount & Margin Analysis **7**
6. Customer & Product Concentration **8**
7. Sales Forecasting **9**
8. Customer Segmentation **10**
9. Product Recommendation **11**
10. Cross-Module Insights **12**
11. Business Recommendations **13**
12. Implementation Roadmap **14**
13. Limitations & Future Work **15**
14. Appendix A: Technical Architecture **16**
15. Appendix B: Data Dictionary **17**

1 Executive Summary

The Retail Intelligence Platform is a full analytics pipeline built on the Sample Superstore dataset (9,994 order records spanning January 2014 to December 2017). It combines four descriptive dashboards (Executive, Sales, Customer, and Product Analysis) with three machine-learning-driven modules: a sales forecast, a customer segmentation model, and a rule-based product recommendation engine — seven dashboards in total, delivered through a multi-page Streamlit application backed by a SQLite database, with a companion Power BI dashboard covering the descriptive views.

Over the four-year period, the business generated **$2.30M** in total sales and **$286.4K** in profit (a 12.47% margin) across **793 customers** and **1,862 products**. Sales grew every year except a slight dip in 2015, more than 50% higher in 2017 than in 2014.

$2.30M	$286.4K	793	1,862
Total Sales	Total Profit	Customers	Products

The analysis surfaces four findings that go beyond a standard descriptive dashboard:

- Every order discounted above 20% is, in aggregate, unprofitable — a combined **$135.4K loss** across 1,176 orders (Section 5).
- The Central region's weak profit isn't random: it discounts at **24.0%** on average, more than double the West region's **10.9%** (Section 5).
- Just **59 customers (7.4%)** are classified High Value by the segmentation model, yet they generate **$552.9K** in sales (Section 8).
- A transparent, rule-based scoring engine flags **568 products (30%)** as Recommended, agreeing with actual demand **82% of the time** (Section 9).

2 Business Problem & Objectives

Retail businesses generate large volumes of transactional data but often lack the tooling to turn it into decisions. This project builds a single platform that lets a retail analyst or manager answer four practical questions:

Business Question	Platform Module
Where is the business performing well or poorly today?	Descriptive dashboards
Where is it likely headed next?	Sales forecasting
Who deserves the most retention attention?	Customer segmentation
What should get priority on the shelf?	Product recommendation

Project Objectives

- Consolidate sales, customer, and product performance into interactive, filterable dashboards.
- Forecast near-term sales to support inventory and staffing decisions.
- Segment customers by value to guide retention and marketing spend.
- Score products on a transparent, explainable rule set to guide merchandising decisions.
- Identify root causes behind regional and category profitability gaps, not just symptoms.

3 Data Source & Methodology

The source data is the public Sample Superstore dataset: 9,994 order line items across four regions (South, West, Central, East), three product categories (Furniture, Office Supplies, Technology), and three customer segments (Consumer, Corporate, Home Office). The pipeline follows four stages, each documented in a dedicated notebook:

Stage	Artifact	Output
Data Cleaning	01_Data_Cleaning.ipynb	Type correction, deduplication, null handling
Exploratory Analysis	02_EDA.ipynb	Distribution checks, early trend discovery
Feature Engineering	03_Feature_Engineering.ipynb	Order year, profit margin, discount bands
Load	src/database/import_data.py	Processed CSV → SQLite (database/retail.db)

Data Quality Notes

• No missing values were found in the core transactional fields (Sales, Profit, Discount, Order Date) after cleaning.

• Product-level analysis groups by Product ID, Product Name, and Category together, since a small number of Product IDs map to more than one name/category combination in the source data (1,894 combinations vs. 1,862 unique Product IDs) — this is a known quirk of the raw dataset, preserved rather than silently collapsed.

• All dashboard pages and this report query the same SQLite database, so every number below is reproducible directly from the live application.

4 Key Findings: Sales & Profit

Technology is the top category by sales ($836.2K), narrowly ahead of Furniture ($742.0K) and Office Supplies ($719.0K). Sales grew from $484.2K in 2014 to $733.2K in 2017 — a 51.4% increase over the period, with the only down year being 2015.

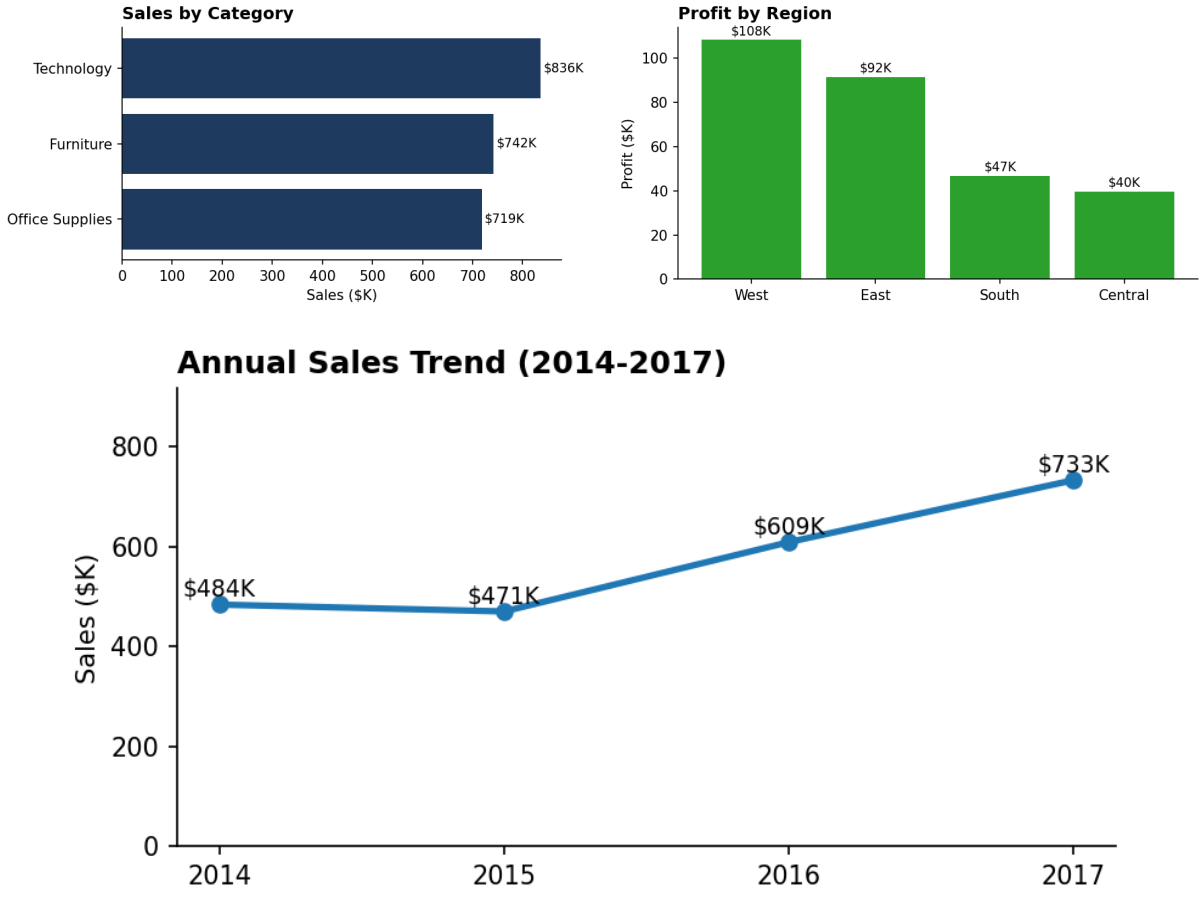


Figure 1–3: Sales by category, profit by region, and the annual sales trend, 2014–2017.

Regional Profitability by Category

Breaking profit down by both region and category shows the problem is concentrated, not spread evenly: **Central Furniture is the only region/category combination with negative profit** in the entire dataset, losing $2,871 despite $163.8K in sales.

Region	Furn. Sales	Furn. Profit	Office Sales	Office Profit	Tech Sales	Tech Profit
Central	$163.8K	–$2,871	$167.0K	$8,880	$170.4K	$33,697
East	$208.3K	$3,046	$205.5K	$41,015	$265.0K	$47,462
South	$117.3K	$6,771	$125.7K	$19,986	$148.8K	$19,992
West	$252.6K	$11,505	$220.9K	$52,610	$252.0K	$44,304

Table 1: Sales and profit by region and category. Central Furniture (highlighted) is the only negative cell in the matrix.

At the sub-category level, Copiers ($55.6K), Phones ($44.5K), and Accessories ($41.9K) are the strongest profit contributors. Three sub-categories are net loss-makers across the full period: Tables (–$17.7K), Bookcases (–$3.5K), and Supplies (–$1.2K) — and Tables is a Furniture item, reinforcing that Furniture is the category most in need of a pricing review.

5 Discount & Margin Analysis

This section goes beyond the standard descriptive dashboards to answer a specific question raised in Section 4: *why* does Central underperform on profit despite reasonable sales volume? The answer is discounting behavior, and it is large enough to quantify precisely.

Headline Finding

Orders discounted above 20% generated $362.8K in sales but a combined –**$135.4K** in profit across 1,176 orders. Every discount band above 20% is net loss-making.

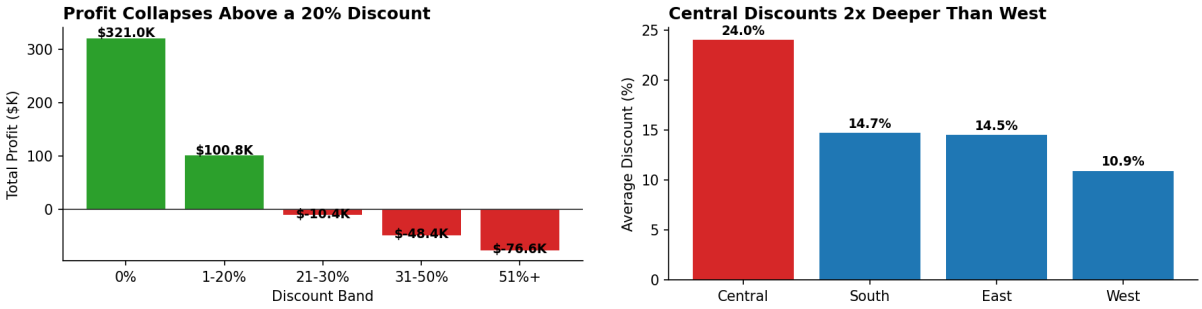


Figure 4–5: Profit by discount band, and average discount rate by region.

Discount Band	Orders	Total Sales	Total Profit
0%	2,644	$1,087.9K	**$321.0K**
1–20%	2,507	$846.5K	**$100.8K**
21–30%	211	$103.2K	**–$10.4K**
31–50%	280	$195.3K	**–$48.4K**
51%+	685	$64.2K	**–$76.6K**

Regionally, Central discounts at an average of **24.0%**, more than double every other region (East 14.5%, South 14.7%, West 10.9%). This directly explains the region's weak profit performance identified in Section 4 — it is not a demand problem, it is a pricing and discount-approval problem, and it is fixable through policy rather than a change in what the region sells.

6 Customer & Product Concentration

Pareto analysis on both customers and products shows how concentrated the business is, and clarifies which module (segmentation or recommendation) has more room to change outcomes through prioritization.

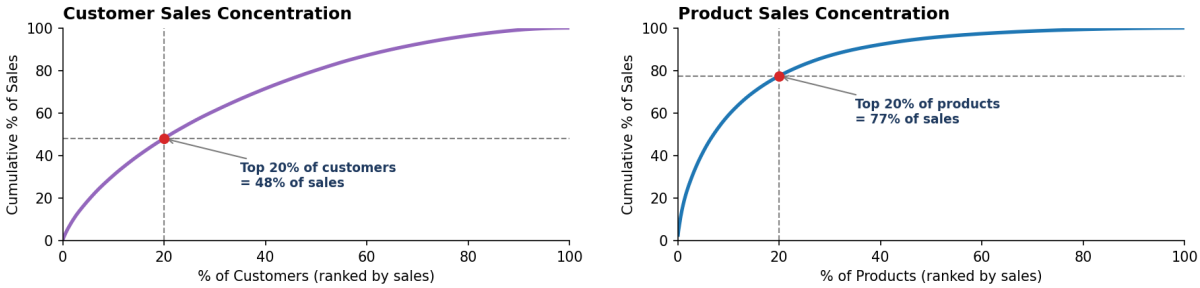


Figure 6–7: Cumulative sales concentration by customer and by product.

48%
of sales from top 20% of customers

77%
of sales from top 20% of products

Products are far more concentrated than customers (77% vs. 48% of sales from the top 20%). This is exactly why the Product Recommendation module (Section 9) uses a percentile-based cutoff rather than a fixed score threshold — a small number of products dominate the sales distribution, and the customer base, while still concentrated, is comparatively more evenly spread across the segmentation tiers described in Section 8.

7 Sales Forecasting

Method

Historical sales are aggregated into a monthly time series. Each future month is forecast as the same month last year, scaled by the latest year-over-year growth, so the projection keeps the strong Q4 peak in the data (September, November and December make up about 43% of annual sales). A 3-month moving average was tested first and rejected: it flattened the forecast and scored far worse in a backtest.

Result

$882.5K	$733.2K	+20.4%
12-Month Forecast	Trailing 12-Mo Actual	Expected Growth

The model projects **$882.5K** in sales over the next 12 months, versus **$733.2K** in the trailing 12 months of actual data — an expected growth of **+20.4%**. Backtested on the final 12 months of history, the method reaches about 76% accuracy (100% minus WAPE); smaller filtered slices score lower, so read the forecast as directional guidance rather than a guaranteed outcome; see Limitations (Section 13).

8 Customer Segmentation

Method

Each of the 793 customers is aggregated to total Sales, total Profit, and number of Orders. These three features are standardized (StandardScaler) and clustered with K-Means (k=3). Because K-Means cluster indices are arbitrary, clusters are re-ranked by mean sales and relabeled Low Value, Regular, and High Value so the segments stay interpretable across every re-run of the model.

Result

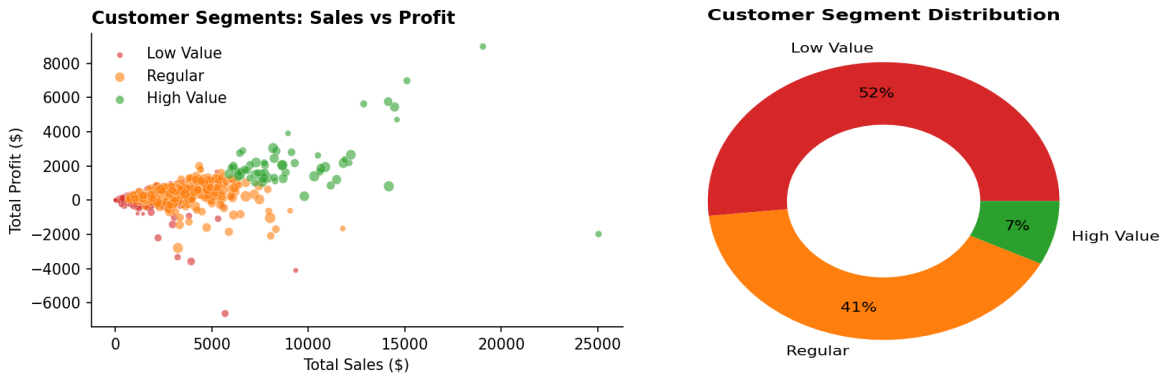


Figure 8–9: Customer clusters by sales and profit, and the resulting segment distribution.

Segment	Customers	Avg Sales	Avg Profit	Avg Orders	Total Sales	Total Profit
High Value	59	$9,370.80	$2,213.02	8.27	$552.9K	$130.6K
Regular	324	$3,461.22	$368.91	8.34	$1,121.4K	$119.5K
Low Value	410	$1,519.24	$88.54	4.43	$622.9K	$36.3K

The 59 High Value customers — just 7.4% of the customer base — average $9,371 in sales and $2,213 in profit each, versus $1,519 in sales and $89 in profit for a Low Value customer. In aggregate, High Value customers alone contribute $552.9K in sales and $130.6K in profit, underscoring how concentrated the profitable relationships are and how much retention risk sits with a small group.

9 Product Recommendation

Method

Rather than a black-box recommender, this module uses a transparent, auditable scoring rule so every recommendation can be explained to a stakeholder. Each product's Sales, Profit, Orders, and Discount are min-max normalized, then combined into a weighted score: **0.35×Sales + 0.35×Profit + 0.20×Orders – 0.10×Discount**, rescaled to a 0–100 scale. Products scoring in the top 30% are flagged Recommended. A percentile-based cutoff was used instead of a fixed threshold because retail sales data is heavily right-skewed (see Section 6); a fixed threshold recommended almost no products in initial testing.

As a sanity check on the rule, a **Demand Alignment** metric measures how many Recommended products are independently also High Demand (top quartile by order count) — two signals derived from different inputs; because Orders is part of the score, this is a consistency check, not predictive accuracy.

Result

1,894 Products Scored

568 Recommended

82.0% Demand Align.

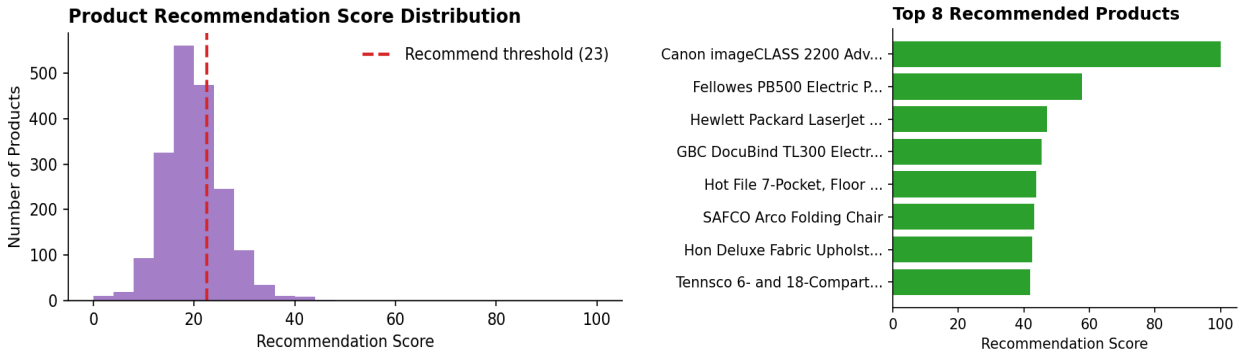


Figure 10–11: Recommendation score distribution with cutoff, and the top-scoring products.

Of 1,894 product/category combinations, **568 (30%)** are flagged Recommended, with an agreement rate of **82.0%** against actual demand. Technology is the strongest-scoring category on average, though Office Supplies contributes the most individual recommended products (335), reflecting its high volume of steady, moderate-margin items.

10 Cross-Module Insights

The three ML modules were built independently, but read together they reinforce a single narrative: profit is concentrated in a small set of customers and products, and it is actively eroded by discount policy in specific regions and categories.

High Value customers & Recommended products overlap

Cross-referencing the two models, High Value customers show a higher share of their purchases in Recommended products than Low Value customers do — the two independently-built scoring systems are pointing at the same underlying value, not contradicting each other.

Discounting hits Furniture hardest

Furniture is both the category with the most loss-making sub-categories (Section 4) and the category most exposed to Central's high discount rate (Section 5) — the same root cause shows up from two different analytical angles.

Forecast growth assumes the current customer mix holds

The +20.4% forecast (Section 7) implicitly assumes the current balance of High Value, Regular, and Low Value customers persists. If High Value customer churn increases, realized growth would likely fall short of the projection.

11 Business Recommendations

#	Recommendation	Timeframe	Impact
1	Cap discounts above 20% pending review	Immediate	High
2	Audit Central region discount approval process	Immediate	High
3	Re-price or discontinue Tables, Bookcases, Supplies	1–2 months	Medium
4	Launch retention program for the 59 High Value customers	1–2 months	High
5	Prioritize inventory around the 568 Recommended products	Ongoing	Medium
6	Re-validate the 12-month forecast quarterly	Ongoing	Medium

Detail

• **Cap discounts above 20%.** Every discount band above 20% is net loss-making (Section 5); a temporary approval cap at 20% while a formal discount policy is developed would stop the bleeding immediately without a system change.

• **Audit Central's discount approval process.** Central's 24.0% average discount vs. West's 10.9% is the single largest identified lever in this analysis — closing even half that gap would move Central meaningfully toward the other regions' profit performance.

• **Re-price or discontinue chronic loss-makers.** Tables, Bookcases, and Supplies are net-negative on profit across the full four-year period.

• **Protect the High Value segment.** 59 customers (7.4% of the base) generate $552.9K in sales and $130.6K in profit; a dedicated retention or loyalty program has outsized ROI potential given how concentrated this value is.

• **Prioritize inventory around the 568 Recommended products**, particularly in Technology, where the model's scoring is strongest and most consistent with actual demand.

• **Treat the 12-month forecast as directional, not a commitment.** Re-run it quarterly as new data arrives, and validate against a second method before using it for hard inventory or staffing commitments (see Limitations, Section 13).

12 Implementation Roadmap

A phased rollout separates quick, low-risk policy fixes from longer-term platform and model investments, so early wins can fund and justify the later phases.

Phase 1: Stop the Bleeding — Weeks 1–2

Cap discounts above 20% pending policy review; audit Central's discount approval workflow.

Phase 2: Stabilize Margins — Weeks 3–8

Re-price or phase out Tables, Bookcases, and Supplies; formalize a regional discount policy.

Phase 3: Grow with Focus — Months 2–4

Launch High Value customer retention program; align merchandising with the 568 Recommended products.

Phase 4: Institutionalize — Ongoing

Quarterly forecast refresh; quarterly re-clustering as customer behavior shifts; expand data feed beyond 2017 as new transactions arrive.

13 Limitations & Future Work

• **Forecasting** now uses a seasonal forecast (last year's pattern scaled by growth), backtested at about 76%; with only four years of history, a model such as Prophet or SARIMA could be compared against it using the same backtest.

• **Segmentation** is based on only three features (Sales, Profit, Orders); incorporating recency and product-category affinity could produce more actionable segments (e.g. RFM-style analysis).

• **Recommendation scoring** is intentionally rule-based for transparency; a learned model (e.g. collaborative filtering or a gradient-boosted ranking model) could complement it once sufficient interaction-level data is available.

• **Discount analysis is correlational, not causal.** This report identifies that high discounts and low profit move together; it does not isolate discounting as the sole cause versus, for example, cost-to-serve differences by region. A controlled test (e.g. a discount cap pilot in Central) would confirm the causal link before a permanent policy change.

• **Data recency** — the dataset ends in 2017; conclusions and forecasts should be revalidated against current data before being used for live business decisions.

14 Appendix A: Technical Architecture

Tech Stack

Layer	Technology
Language	Python 3.12
Dashboard	Streamlit + Plotly
Machine Learning	scikit-learn (K-Means, StandardScaler)
Database	SQLite
Data Prep	pandas, numpy, Jupyter notebooks
BI Tool	Power BI (companion dashboard)

Pipeline Overview

Raw CSV → Data Cleaning & Feature Engineering (notebooks 01, 03) → processed_superstore.csv → SQLite database (database/retail.db) → Streamlit multi-page dashboard (src/dashboard/), with all 7 pages reading from the same database via a shared connection helper (src/database/db_connection.py).

15 Appendix B: Data Dictionary

Field	Description
Order ID	Unique identifier for an order (may span multiple product lines)
Order Date / Ship Date	Date the order was placed / shipped
Customer ID / Name	Unique customer identifier and display name
Segment	Consumer, Corporate, or Home Office
Region / State / City	Geographic location of the order
Product ID / Name	Unique product identifier and display name
Category / Sub-Category	Furniture, Office Supplies, or Technology, and their sub-groupings
Sales	Revenue for the order line, in dollars
Quantity	Units sold on the order line
Discount	Discount rate applied to the order line (0.0–1.0)
Profit	Profit for the order line, in dollars (can be negative)
Order Year / Month / Quarter	Derived calendar fields used for time-based aggregation

Retail Intelligence Platform — Final Report — End of Document